# Distributional Forecasting of FX Volatility with Aleatoric and Epistemic Uncertainty

Mehmet Kurmaz

April 2026

**Abstract**

Forecasting FX volatility as a distribution rather than a point estimate is more useful for risk management because tail risk and forecast uncertainty matter directly for decisions such as VaR, expected shortfall, and hedging. This paper studies daily USD/MXN volatility using a probabilistic patch-based Transformer that maps the previous 40 trading days of multivariate FX data into a Gaussian forecast for next-day log-volatility. A deep ensemble separates aleatoric from epistemic uncertainty. Using daily data from 2000–2026, the model is compared with HAR-style, GARCH, GJR-GARCH, EGARCH, and quantile-HAR benchmarks. The Transformer is competitive and improves over a univariate version when cross-currency information is included, but classical models remain stronger overall, with EGARCH achieving the best density forecast performance. Still, the Transformer provides well-calibrated intervals and informative epistemic-uncertainty spikes during regime shifts, suggesting value as a flexible framework for richer data and explicit uncertainty management.

**Keywords:** foreign exchange volatility, probabilistic forecasting, Transformer, uncertainty quantification, risk management.

## 1 Introduction

Volatility forecasts are central to foreign-exchange (FX) risk management. Treasury desks, macro hedge funds, and corporates with currency exposure all need a view on future variability in order to size positions, set margin buffers, hedge cash-flow risk, and quantify downside scenarios. A point forecast alone is not enough for these tasks. Two models can produce the same expected volatility while implying very different tail risk, capital usage, and stress losses. Distributional forecasting therefore matters directly for value-at-risk and expected-shortfall style decisions (Jorion, 2007; McNeil et al., 2015; Diebold et al., 1998; Gneiting and Raftery, 2007). In the same spirit, uncertainty quantification matters because forecast risk comes from at least two distinct sources, the market itself is intrinsically noisy (aleatoric uncertainty), and the model can also be unsure because the data are sparse, the regime changes, or the specification is incomplete (epistemic uncertainty) (Kendall and Gal, 2017; Lakshminarayanan et al., 2017).

This paper studies next-day volatility forecasting for USD/MXN. The pair is economically important because the Mexican peso is one of the most actively traded emerging-market currencies and a large share of its turnover takes place offshore, making it a useful laboratory for studying liquid but macro-sensitive FX risk (Banco de Mexico, 2022). The empirical task is intentionally difficult. Only daily spot exchange-rate data are used, so the volatility target must be proxied by squared daily returns rather than by high-frequency realized volatility. This setup is realistic for many risk-management environments where intraday data, option-implied information, or full news feeds are unavailable.

The paper proposes a probabilistic patch-based Transformer for the conditional distribution of next-day USD/MXN log-volatility. The idea is inspired by modern sequence models for time series (Lim et al., 2021; Nie et al., 2023; Wen et al., 2023) and by recent FX-specific Transformer work (Zhao and Yan, 2024; Nguyen et al., 2025), but the design here is deliberately compact. The previous 40 trading days are split into non-overlapping patches, embedded, processed by self-attention, and mapped to the parameters of a Gaussian predictive density. A deep ensemble of such models provides a clean decomposition of predictive uncertainty into aleatoric and epistemic components.

Formally, the object of interest is the conditional law

$$p(y_{t+1} \mid \mathcal{F}_t), \tag{1}$$

where $y_{t+1}$ is next-day log-volatility and $\mathcal{F}_t$ is the information set generated by all observed covariates up to day $t$. The contribution of the paper is therefore threefold, but the logic is easier to state in one line than as a checklist, the forecasting target is treated as a conditional *distribution* rather than a point target; a compact patch-based Transformer is parameterized to output $(\mu_t, \sigma_t^2)$ and then aggregated through an ensemble-based uncertainty decomposition; and the resulting density forecasts are benchmarked against strong classical competitors, namely Gaussian HAR-style dynamics, GARCH, GJR-GARCH, EGARCH, and quantile-HAR.

The empirical conclusion is negative but useful. On this daily proxy target, low-dimensional classical models remain stronger on NLL, CRPS, and QLIKE. However, the Transformer still learns sensible volatility regimes, benefits from cross-currency features relative to a univariate version, and produces an interpretable model-risk signal through epistemic uncertainty.

## 2 A Short Background Review

This section reviews only the model classes that are estimated later, and it fixes the notation used throughout the paper. Let $y_t$ denote the transformed volatility target and let $\mathcal{F}_{t-1}$ be the information set available at the close of day $t-1$. A one-step-ahead forecaster can target the conditional mean $\mathbb{E}[y_t \mid \mathcal{F}_{t-1}]$, a conditional quantile $Q_q(y_t \mid \mathcal{F}_{t-1})$, or the full predictive density $p(y_t \mid \mathcal{F}_{t-1})$. The benchmark set used in this paper deliberately spans all three objects.

A natural starting point is the heterogeneous autoregressive idea of Corsi (2009). On the transformed daily target, a HAR-style mean specification takes the form

$$y_t = \beta_0 + \beta_d y_{t-1} + \beta_w \bar{y}_{t-1}^{(5)} + \beta_m \bar{y}_{t-1}^{(22)} + e_t, \tag{2}$$

where

$$\bar{y}_{t-1}^{(5)} = \frac{1}{5}\sum_{j=1}^{5} y_{t-j}, \qquad \bar{y}_{t-1}^{(22)} = \frac{1}{22}\sum_{j=1}^{22} y_{t-j}, \tag{3}$$

and $e_t$ is an innovation term. The attraction of (2) is that it hard-codes volatility persistence at daily, weekly, and monthly horizons without requiring a large parameter vector. In the Gaussian version used later, $e_t \sim \mathcal{N}(0, \sigma^2)$, so the model delivers a full density as well as a point forecast.

A second family models the dynamics of the conditional variance itself. Writing

$$e_t = \sqrt{h_t}\, z_t, \qquad z_t \overset{i.i.d.}{\sim} \mathcal{N}(0,1), \tag{4}$$

The standard GARCH(1, 1) recursion, GJR-GARCH specification of Glosten et al. (1993) and EGARCH (Nelson, 1991) are outlined more in detail in Subsection 4.1.

These models are especially relevant because they provide parsimonious benchmarks for volatility clustering and leverage-type asymmetry (Engle, 1982; Bollerslev, 1986; Nelson, 1991; Glosten et al., 1993; Hansen and Lunde, 2005). In the present study they are applied to the residual process of the HAR-style mean equation, so that the comparison is between density models defined on the same transformed target.

The quantile-HAR model targets interval structure directly rather than committing to a parametric shape. For any quantile level $q \in (0, 1)$,

$$Q_q(y_t \mid \mathcal{F}_{t-1}) = \alpha_q + \beta_{d,q} y_{t-1} + \beta_{w,q}\bar{y}_{t-1}^{(5)} + \beta_{m,q}\bar{y}_{t-1}^{(22)}, \tag{5}$$

with coefficients estimated by minimizing the pinball loss (Koenker and Bassett, 1978; Koenker, 2005). This benchmark is useful because it separates interval forecasting from the Gaussian assumption imposed by the density models.

The paper also distinguishes two sources of predictive uncertainty. If $\Theta$ denotes a random model instance, for example one draw from an ensemble of independently trained neural networks, then the law of total variance gives

$$\mathbb{V}\mathrm{ar}(Y_{t+1} \mid \mathcal{F}_t) = \underbrace{\mathbb{E}_\Theta[\mathbb{V}\mathrm{ar}(Y_{t+1} \mid \mathcal{F}_t, \Theta)]}_{U_t^{\mathrm{ale}}} + \underbrace{\mathbb{V}\mathrm{ar}_\Theta(\mathbb{E}[Y_{t+1} \mid \mathcal{F}_t, \Theta])}_{U_t^{\mathrm{epi}}}. \tag{6}$$

The first term, $U_t^{\mathrm{ale}}$, is aleatoric uncertainty which measures the irreducible randomness of the market conditional on the model. The second term, $U_t^{\mathrm{epi}}$, is epistemic uncertainty which measures model disagreement and therefore falls, at least in principle, as the specification and data become more informative (Kendall and Gal, 2017; Lakshminarayanan et al., 2017). For an ensemble of Gaussian predictors

$$p_m(y_{t+1} \mid \mathcal{F}_t) = \mathcal{N}(\mu_{m,t}, \sigma_{m,t}^2), \qquad m = 1, \ldots, M, \tag{7}$$

equation (6) becomes

$$U_t^{\mathrm{ale}} = \frac{1}{M}\sum_{m=1}^{M} \sigma_{m,t}^2, \qquad U_t^{\mathrm{epi}} = \frac{1}{M}\sum_{m=1}^{M} (\mu_{m,t} - \bar{\mu}_t)^2, \qquad \bar{\mu}_t = \frac{1}{M}\sum_{m=1}^{M} \mu_{m,t}. \tag{8}$$

This decomposition is central to the interpretation of the proposed Transformer later in the paper.

# 3 Problem Setup, Data, and Feature Set

## 3.1 Forecasting object and notation

Let $S_t$ denote the USD/MXN spot rate at business day $t$, quoted as MXN per USD. Returns are defined by

$$r_t = \Delta \log S_t = \log S_t - \log S_{t-1}, \tag{9}$$

and the one-day volatility proxy is

$$RV_t = r_t^2. \tag{10}$$

Because $RV_t$ is strictly positive and extremely right-skewed, the working target is the log proxy

$$y_t = \log(RV_t + \varepsilon), \qquad \varepsilon = 10^{-8}. \tag{11}$$

For each day let $x_t \in \mathbb{R}^{23}$ denote the feature vector that stacks cross-currency returns, cross-currency log-volatility proxies, calendar dummies, and HAR-style aggregates. With filtration

$$\mathcal{F}_t = \sigma(x_1, \ldots, x_t), \tag{12}$$

the statistical task is to estimate the full one-step-ahead density

$$p(y_{t+1} \mid \mathcal{F}_t), \tag{13}$$

and not only its conditional mean. On the original scale, the forecasted object is the positive random variable $RV_{t+1} = \exp(y_{t+1}) - \varepsilon$.

## 3.2 Data source and sample

The core series is the daily Mexican peso per U.S. dollar spot rate (FRED series DEXMXUS) (Federal Reserve Bank of St. Louis, 2026), accessed through the public `datasets/exchange-rates` mirror (datasets/exchange-rates contributors, 2026). To inject multivariate FX context, the synchronous daily spot rates of seven additional currencies are included: Brazil, Canada, Euro area, South Africa, Japan, South Korea, and China. After aligning dates and removing missing observations, the final panel spans 2000-01-03 to 2026-03-27 and contains 6,577 common business-day observations.

Equation (11) does not produce true realized volatility, because only close-to-close daily spot data are available. The target is therefore a deliberately noisy proxy. This matters empirically, if $y_t$ has low signal-to-noise ratio, parsimonious models that exploit persistence can remain difficult to beat. At the same time, the setup is operationally realistic for institutions that do not maintain an intraday data stack or an option-implied volatility surface for every currency pair they monitor.

## 3.3 Feature engineering and sample split

Each day contributes a 23-dimensional covariate vector. For each of the eight currencies, the daily return and the daily log-volatility proxy are included, yielding 16 market features. These are augmented with five day-of-week dummies. In addition, the model receives two HAR-style

aggregates, $\bar{y}_t^{(5)} = \frac{1}{5}\sum_{j=0}^{4} y_{t-j}$ and $\bar{y}_t^{(22)} = \frac{1}{22}\sum_{j=0}^{21} y_{t-j}$, which expose both high-frequency shocks and lower-frequency persistence. The data are split chronologically into training (up to 2015-12-31), validation (2016-01-01 to 2019-12-31), and test (2020-01-01 onward) periods. This design places the COVID episode, the 2022 tightening cycle, and the post-2023 sample fully out of sample.

**Figure 1:** USD/MXN daily spot rate over the full sample. Dashed vertical lines mark the training/validation and validation/test splits. The series exhibits several regime changes, which is precisely the setting in which time-varying volatility and uncertainty quantification become economically relevant.

Figure 2 adds two descriptive diagnostics that motivate the model comparison. Panel (a) shows that the training-sample autocorrelation function of $y_t$ is significantly positive across a long range of short lags, which is exactly the environment in which HAR and GARCH-type persistence structures should be strong. Panel (b) constructs an exploratory directed network of lagged cross-currency linkages. For every ordered pair $(i, j)$, the edge weight is

$$w_{i\to j} = \Delta R^2_{i\to j}\,\mathbf{1}\{p_{ij} < 10^{-3}\}, \qquad \Delta R^2_{i\to j} = R^2\Big(y_t^{(j)} \sim y_{t-1}^{(j)}, y_{t-1}^{(i)}\Big) - R^2\Big(y_t^{(j)} \sim y_{t-1}^{(j)}\Big), \quad (14)$$

where $p_{ij}$ is the single-restriction $F$-test $p$-value for the additional lagged regressor. This is not a causal claim; it is a descriptive lagged dependence map. Even so, the network shows that USD/MXN is not isolated, which is precisely the empirical motivation for testing a multivariate Transformer against a univariate alternative.

# 4 Models

## 4.1 Classical baselines

The first baseline is a Gaussian HAR-style model for the transformed target. Let

$$\bar{y}_{t-1}^{(5)} = \frac{1}{5}\sum_{j=1}^{5} y_{t-j}, \qquad \bar{y}_{t-1}^{(22)} = \frac{1}{22}\sum_{j=1}^{22} y_{t-j}. \quad (15)$$

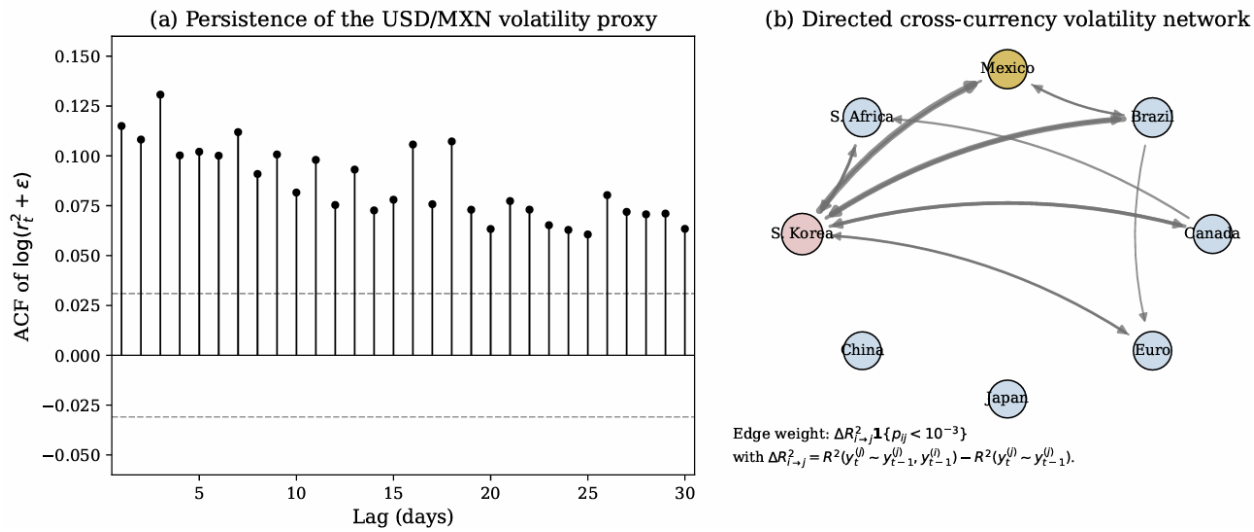



**Figure 2:** Descriptive structure of the training sample. Panel (a) shows pronounced short-run persistence in the USD/MXN log-volatility proxy. Panel (b) visualizes significant lagged cross-currency predictive linkages using the exploratory edge weight in (14). Mexico is highlighted to emphasize the target market; South Korea and Brazil emerge as especially connected nodes in this simple lagged network.

Then the conditional mean is

$$y_t = \beta_0 + \beta_d y_{t-1} + \beta_w \bar{y}_{t-1}^{(5)} + \beta_m \bar{y}_{t-1}^{(22)} + e_t, \qquad e_t \sim \mathcal{N}(0, \sigma^2). \tag{16}$$

This is a direct analogue of HAR-RV adapted to the daily log squared-return proxy.

The second baseline is a quantile-HAR model. For each quantile level $q \in \{0.05, 0.50, 0.95\}$,

$$Q_q(y_t \mid \mathcal{F}_{t-1}) = \alpha_q + \beta_{d,q} y_{t-1} + \beta_{w,q} \bar{y}_{t-1}^{(5)} + \beta_{m,q} \bar{y}_{t-1}^{(22)}, \tag{17}$$

with coefficients estimated by quantile regression (Koenker and Bassett, 1978; Koenker, 2005). This baseline directly targets interval prediction without imposing a parametric Gaussian shape.

The third family combines the HAR conditional mean with conditional heteroskedastic residuals. Let $m_t$ denote the fitted mean from (16), and define $e_t = y_t - m_t$. Then:

$$\text{GARCH:} \quad h_t = \omega + \alpha e_{t-1}^2 + \beta h_{t-1}, \tag{18}$$

$$\text{GJR-GARCH:} \quad h_t = \omega + \alpha e_{t-1}^2 + \gamma \mathbf{1}\{e_{t-1} < 0\} e_{t-1}^2 + \beta h_{t-1}, \tag{19}$$

$$\text{EGARCH:} \quad \log h_t = \omega + \beta \log h_{t-1} + \alpha(|z_{t-1}| - \mathbb{E}|z|) + \gamma z_{t-1}, \qquad z_t = e_t/\sqrt{h_t}. \tag{20}$$

Conditional on $\mathcal{F}_{t-1}$, each baseline implies

$$y_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(m_t, h_t). \tag{21}$$

Strictly speaking, GARCH models were designed for return innovation variance rather than for the transformed log-volatility target used here. In this report they are used as strong, low-dimensional conditional-density benchmarks on the same target and loss function.

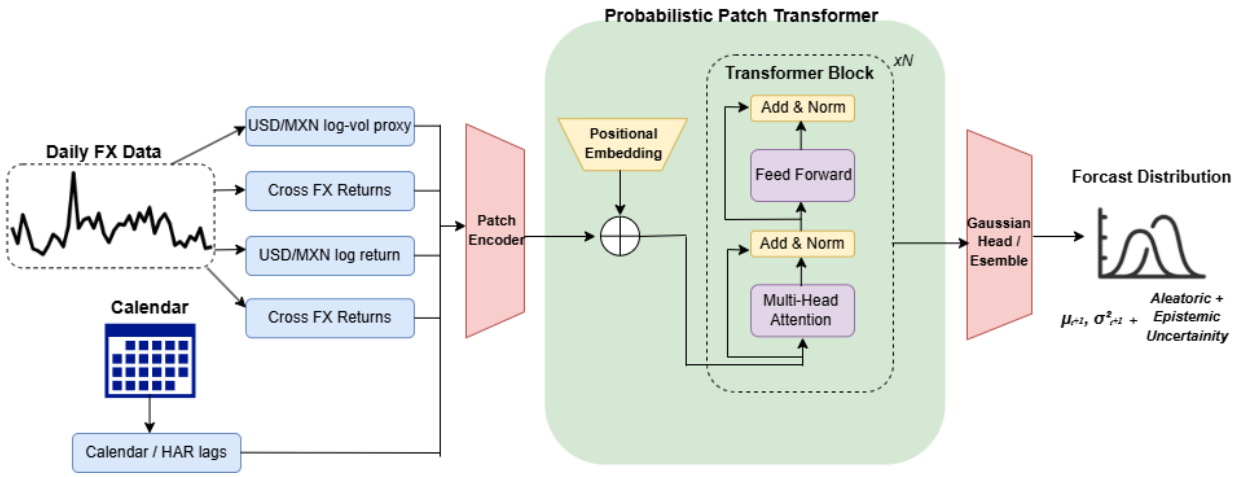



**Figure 3:** Proposed probabilistic forecasting framework. Past daily FX returns, volatility proxies, and calendar/HAR features are converted into patch tokens, processed by a compact Transformer, and mapped to a Gaussian predictive density. Multiple independently trained models form a deep ensemble, enabling an explicit decomposition into aleatoric and epistemic uncertainty.

## 4.2 Probabilistic patch-based Transformer

Let $X_{t-L:t-1} \in \mathbb{R}^{L\times d}$ denote the matrix of the previous $L = 40$ daily feature vectors, with $d = 23$. The model first partitions the lookback window into $P = L/5 = 8$ non-overlapping patches of length 5. Patch $i$ is flattened into

$$p_i = \mathrm{vec}(X_{t-40+5(i-1):t-40+5i-1}) \in \mathbb{R}^{5d}. \tag{22}$$

Each patch is linearly projected into the model dimension,

$$z_i = W_p p_i + b_p + e_i^{\mathrm{pos}}, \tag{23}$$

where $e_i^{\mathrm{pos}}$ is a learnable positional embedding. A learned class token $z_{\mathrm{cls}}$ is prepended, and the resulting token sequence is passed through a Transformer encoder. Let $h_t$ denote the final class-token representation. The probabilistic head outputs the parameters of a Gaussian density,

$$(\mu_t, \log \sigma_t^2) = g_\theta(h_t), \qquad y_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(\mu_t, \sigma_t^2). \tag{24}$$

The architecture is intentionally compact, one encoder layer, four heads, model width 16, feed-forward width 32, and strong early stopping. The model is a small sequence model trained from scratch for the conditional distribution of next-day log-volatility. Figure 3 summarizes the framework.

## 4.3 Likelihood and uncertainty decomposition

Training minimizes the Gaussian negative log-likelihood (NLL),

$$\mathcal{L}_{\mathrm{NLL}}(\theta) = \frac{1}{N}\sum_{t=1}^{N}\left[\frac{1}{2}\log(2\pi\sigma_t^2) + \frac{(y_t-\mu_t)^2}{2\sigma_t^2}\right]. \tag{25}$$

Because the model is trained on the log-volatility target, the implied conditional mean of the original-scale proxy is

$$\mathbb{E}[RV_t \mid \mathcal{F}_{t-1}] = \exp\left(\mu_t + \frac{1}{2}\sigma_t^2\right) - \varepsilon. \tag{26}$$

To quantify uncertainty beyond a single fit, an ensemble of $M = 3$ independently initialized Transformers is trained. The ensemble predictive law is the equally weighted Gaussian mixture

$$p_{\text{ens}}(y_t \mid \mathcal{F}_{t-1}) = \frac{1}{M}\sum_{m=1}^{M} \mathcal{N}\Big(y_t; \mu_{m,t}, \sigma_{m,t}^2\Big). \tag{27}$$

For reporting and scoring, this mixture is moment-matched to a Gaussian with mean

$$\bar{\mu}_t = \frac{1}{M}\sum_{m=1}^{M} \mu_{m,t} \tag{28}$$

and variance

$$\bar{\sigma}_t^2 = \underbrace{\frac{1}{M}\sum_{m=1}^{M} \sigma_{m,t}^2}_{U_t^{\text{ale}}} + \underbrace{\frac{1}{M}\sum_{m=1}^{M} (\mu_{m,t} - \bar{\mu}_t)^2}_{U_t^{\text{epi}}}. \tag{29}$$

Equation (29) is the law of total variance specialized to the ensemble. The first term, $U_t^{\text{ale}}$, is within-model uncertainty and reflects irreducible market noise. The second term, $U_t^{\text{epi}}$, is between-model disagreement and therefore acts as a model-risk indicator. In a risk-management setting those two warnings are different, large $U_t^{\text{ale}}$ calls for wider volatility bands, while large $U_t^{\text{epi}}$ suggests less confidence in the specification itself.

# 5 Experimental Design and Evaluation

All models are estimated on the training split only. The validation split is reserved for early stopping and architecture selection for the Transformer, and the test split begins in 2020. Forecasts are strictly one-step-ahead and chronological, so no look-ahead leakage is possible. For each model let $F_t$ denote the forecast CDF for $y_t$ formed at time $t-1$, with predictive mean $\hat{\mu}_t$ and, when relevant, predictive interval $[l_{t,\alpha}, u_{t,\alpha}]$.

The primary scoring rule is the negative log-likelihood,

$$\ell_t^{\text{NLL}} = -\log f_t(y_t), \tag{30}$$

because the forecasting object in (13) is a conditional density. To avoid evaluating density forecasts with one score only, the report also uses the continuous ranked probability score,

$$\text{CRPS}(F_t, y_t) = \int_{-\infty}^{\infty} \left(F_t(z) - \mathbf{1}\{y_t \le z\}\right)^2 \, dz, \tag{31}$$

the central $(1-\alpha)$ interval score,

$$\text{IS}_\alpha(l, u; y) = (u - l) + \frac{2}{\alpha}(l - y)\mathbf{1}\{y < l\} + \frac{2}{\alpha}(y - u)\mathbf{1}\{y > u\}, \tag{32}$$

and the empirical coverage

$$\mathrm{Cov}_{1-\alpha} = \frac{1}{T_{\text{test}}} \sum_t \mathbf{1}\{l_{t,\alpha} \le y_t \le u_{t,\alpha}\}. \tag{33}$$

Point diagnostics are reported on the transformed scale through

$$\mathrm{RMSE} = \left( \frac{1}{T_{\text{test}}} \sum_t (y_t - \hat{\mu}_t)^2 \right)^{1/2}, \qquad \mathrm{MAE} = \frac{1}{T_{\text{test}}} \sum_t |y_t - \hat{\mu}_t|. \tag{34}$$

On the original positive scale, the proxy forecast is $\widehat{RV}_t = \exp(\hat{\mu}_t + \hat{\sigma}_t^2/2) - \varepsilon$, and the volatility-oriented loss is

$$\mathrm{QLIKE}(RV_t, \widehat{RV}_t) = \frac{RV_t}{\widehat{RV}_t} - \log\left( \frac{RV_t}{\widehat{RV}_t} \right) - 1, \tag{35}$$

which is standard in volatility work precisely because it is robust to noisy volatility proxies (Patton, 2011). Density calibration is checked visually through the probability integral transform

$$u_t = F_t(y_t), \tag{36}$$

which should be i.i.d. Uniform$(0, 1)$ under correct specification.

Pairwise predictive-accuracy comparisons are summarized with a one-step Diebold-Mariano style statistic. If $d_t = \ell_t^{(A)} - \ell_t^{(B)}$ denotes the difference between per-observation NLLs, then

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\widehat{\mathbb{V}\mathrm{ar}}(d_t)/T_{\text{test}}}}, \qquad \bar{d} = \frac{1}{T_{\text{test}}} \sum_t d_t. \tag{37}$$

Because the horizon is one day, a contemporaneous variance estimator is used. The goal is modest, not to claim asymptotic perfection, but to test whether observed differences in density scores are materially larger than sampling noise.

# 6 Empirical Results

## 6.1 Main accuracy comparison

Table 1 reports the density-forecast results. The headline result is that classical baselines remain stronger. EGARCH attains the best test NLL (2.229), while HAR-style Gaussian is best on CRPS, QLIKE, and interval score among the parametric density models. GARCH and GJR-GARCH are nearly indistinguishable in this application. The univariate Transformer is weaker than all classical baselines. Adding cross-currency information and ensembling improves the Transformer modestly, NLL falls from 2.246 to 2.243, CRPS from 1.275 to 1.265, and MAE from 1.801 to 1.774, but the gap to the classical models remains.

Figure 4 turns Table 1 into a direct visual comparison. The ranking is stable across the sharpness-oriented scores, EGARCH is best on NLL, the HAR-style Gaussian is best on CRPS and QLIKE, and the Transformer ensemble remains behind the classical baselines even

**Table 1:** Out-of-sample density forecast accuracy on the test set. Lower is better, except coverage where closeness to the nominal 0.900 is preferred. Bold indicates the best entry in a column among the density models.

| Model | NLL | CRPS | $\text{RMSE}_y$ | $\text{MAE}_y$ | Cov90 | IS90 | QLIKE |
|---|---|---|---|---|---|---|---|
| HAR-style Gaussian | 2.230 | **1.252** | **2.245** | **1.764** | 0.895 | **9.691** | **1.737** |
| GARCH | 2.230 | 1.252 | **2.245** | **1.764** | 0.899 | 9.723 | 1.763 |
| GJR-GARCH | 2.230 | 1.252 | **2.245** | **1.764** | 0.899 | 9.723 | 1.763 |
| EGARCH | **2.229** | 1.252 | **2.245** | **1.764** | **0.902** | 9.735 | 1.806 |
| Transformer (univariate) | 2.246 | 1.275 | 2.276 | 1.801 | 0.893 | 9.827 | 1.814 |
| Transformer ensemble | 2.243 | 1.265 | 2.261 | 1.774 | 0.907 | 9.862 | 1.921 |

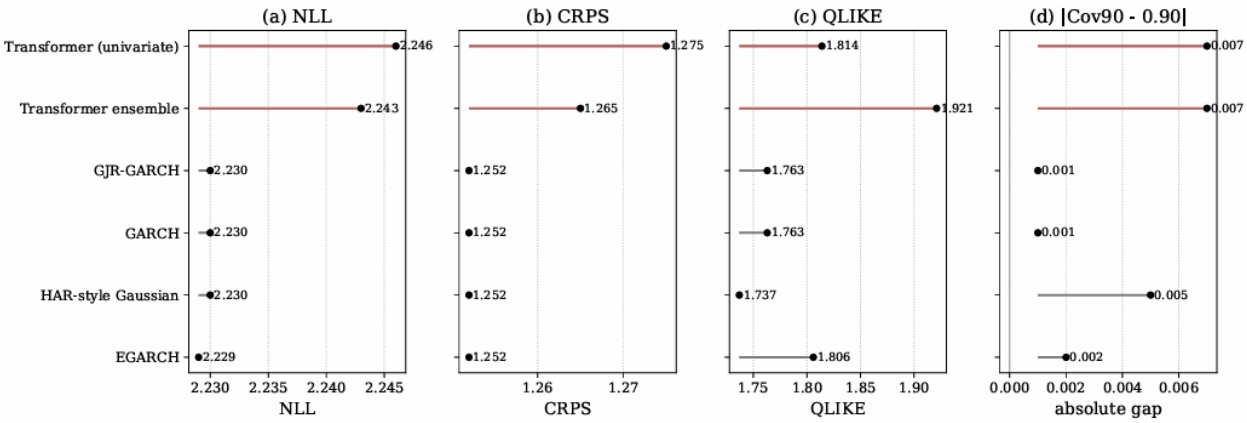



**Figure 4:** Visual summary of the density-score comparison. Lower is better in all four panels. The two Transformer variants improve from univariate to multivariate, but the classical baselines remain tighter on NLL, CRPS, and QLIKE, while the coverage-gap panel shows that GARCH-type models also stay closer to the nominal 90% interval coverage.

after the gain from multivariate conditioning. On the daily squared-return proxy, HAR-style Gaussian and EGARCH outperform the Transformer on NLL, CRPS, and QLIKE, suggesting that low-dimensional persistence dominates the forecastable signal in this setting. This is consistent with the cautionary message of Zeng et al. (2023): complex sequence models do not automatically outperform simpler baselines in time-series forecasting.

Two further experiments make the ranking more transparent. First, define the one-step density-loss differential against benchmark $m$ by

$$\Delta_t^{(m)} = \ell_{t,\text{ens}}^{\text{NLL}} - \ell_{t,m}^{\text{NLL}}, \qquad m \in \{\text{HAR}, \text{EGARCH}\}, \tag{38}$$

so that $\Delta_t^{(m)} < 0$ means the Transformer ensemble is locally better. Figure 5 reports a 60-day rolling average of this statistic. The ensemble is not uniformly dominated, there are short windows in which the loss differential becomes negative. However, the line spends more time above zero than below it, which is consistent with the average NLL ranking in Table 1.

Second, the test set can be stratified by realized volatility regime. Let the realized-scale proxy $RV_t$ be sorted into quintiles, and define

$$\bar{\ell}_q^{(m)} = \frac{1}{|I_q|} \sum_{t \in I_q} \ell_{t,m}^{\text{NLL}}, \tag{39}$$

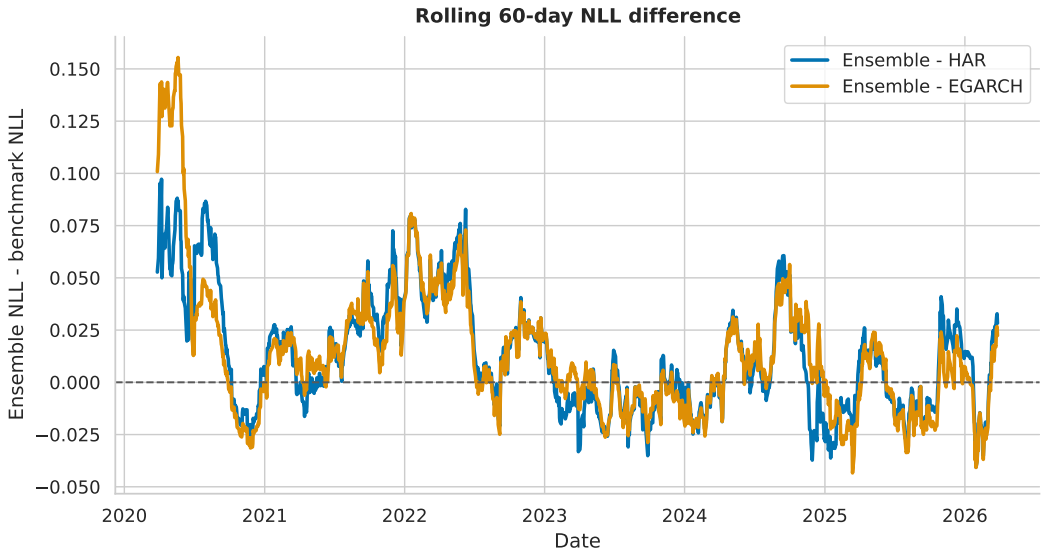



**Figure 5:** Rolling 60-day density-loss differential, defined as ensemble NLL minus benchmark NLL. Values below zero indicate local periods in which the Transformer ensemble outperforms the comparator. The ensemble occasionally improves on HAR-style Gaussian and EGARCH, but the differential is positive more often than negative, which is consistent with the test-set averages.

where $I_q$ is the set of test dates whose realized proxy lies in quintile $q$. Figure 9 shows that the classical models retain their advantage across calm and stressed regimes, although the gaps are smallest in the middle of the distribution. This matters for risk management because it shows that the negative headline result is not driven by a single exceptional period, it is visible across several volatility environments.

Table 2 sharpens the interpretation. Quantile-HAR achieves a lower 90% interval score than the Gaussian density models, but it does so with only 86.6% empirical coverage rather than the nominal 90%. In other words, it buys sharpness partly by issuing intervals that are too narrow. Meanwhile, the Diebold-Mariano comparisons show that the Transformer ensemble underperforms both HAR-style Gaussian and EGARCH on NLL at conventional significance levels. The negative result is therefore statistically meaningful, not just numerically tiny.

## 6.2 What the Transformer learns well

Although the Transformer does not win the horse race, it still exhibits two desirable properties. First, the multivariate ensemble improves on the univariate Transformer on several scores, which implies that conditioning on the broader $\mathcal{F}_t$ rather than a USD/MXN-only subset is empirically worthwhile. Second, its interval calibration is reasonable. Figure 8b shows that empirical coverage tracks nominal coverage closely across a range of confidence levels; at the 90% level, $\mathrm{Cov}_{0.90} = 0.907$. From a risk-management perspective that mild conservatism is preferable to systematic undercoverage, even if it comes with wider intervals than the sharpest classical alternatives.

Figure 6 shows the out-of-sample predictive mean together with the realized target. The predictive mean is much smoother than the realized series, which is unsurprising because

**Table 2:** Interval-focused comparison for the quantile-HAR baseline and pairwise NLL tests for the Transformer ensemble. The quantile model produces relatively sharp intervals but materially undercovers. Negative Diebold-Mariano statistics indicate that the Transformer ensemble has higher NLL than the comparator.

| Model / test | Cov90 | IS90 | $\text{RMSE}_{0.50}$ | $\text{Pinball}_{0.05}$ | $\text{Pinball}_{0.50}$ | $\text{Pinball}_{0.95}$ |
|---|---|---|---|---|---|---|
| Quantile-HAR | 0.866 | 9.470 | 2.253 | 0.295 | 0.854 | 0.179 |

| NLL comparison | DM statistic | $p$-value |
|---|---|---|
| Ensemble vs HAR-style Gaussian | -2.456 | 0.014 |
| Ensemble vs EGARCH | -2.953 | 0.003 |

**Table 3:** Average uncertainty on the test set by calendar year. Values are standard deviations on the log-volatility scale.

| Year | Aleatoric s.d. | Epistemic s.d. |
|---|---|---|
| 2020 | 2.261 | 0.170 |
| 2021 | 2.178 | 0.151 |
| 2022 | 2.298 | 0.184 |
| 2023 | 2.261 | 0.164 |
| 2024 | 2.181 | 0.157 |
| 2025 | 2.192 | 0.125 |
| 2026 | 2.113 | 0.123 |

one-day squared returns are extremely noisy. The model therefore captures broad volatility regimes rather than individual daily spikes, which is the more realistic objective at a daily horizon.

## 6.3 Aleatoric versus epistemic uncertainty

The most distinctive output of the proposed framework is the uncertainty decomposition in (29). Figure 7 shows the full test-period decomposition and a zoom during 2020. Aleatoric uncertainty dominates total predictive uncertainty throughout the sample. This is expected, daily FX volatility is genuinely noisy, and the transformed squared-return proxy amplifies that noise. Epistemic uncertainty is much smaller on average, but it is far from constant. It spikes during periods in which the ensemble members disagree more strongly, which is exactly the behavior desired from a model-risk indicator.

A year-by-year summary appears in Table 3. Average epistemic uncertainty is highest in 2022 (0.184) and also elevated in 2020 (0.170), before drifting lower in 2025-2026. This is sensible. From a risk-management perspective, large aleatoric uncertainty says "the market is noisy," whereas large epistemic uncertainty says "the model itself is less certain." Those are different warnings, and they should trigger different actions. The former may justify wider hedging bands or larger capital buffers; the latter may justify human review, scenario overlays, or model-combination strategies.

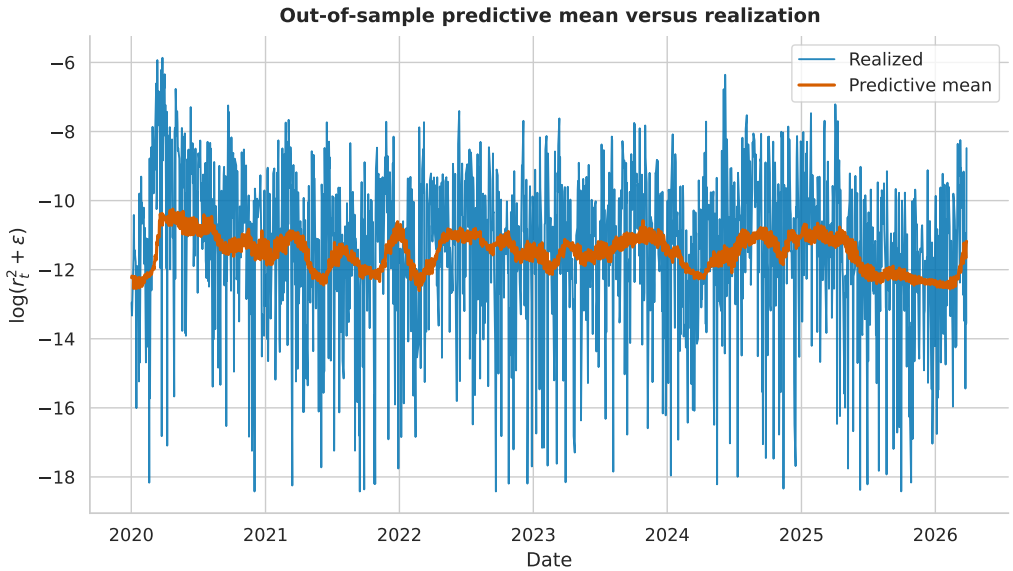



**Figure 6:** Predictive mean versus realized log-volatility proxy. The ensemble tracks medium-run volatility regimes but intentionally smooths many day-to-day spikes, which reflects both the difficulty of the target and the fact that one-step-ahead density forecasting at daily frequency is dominated by persistent rather than idiosyncratic structure.

# 7 Discussion

## 7.1 Why the classical models win here

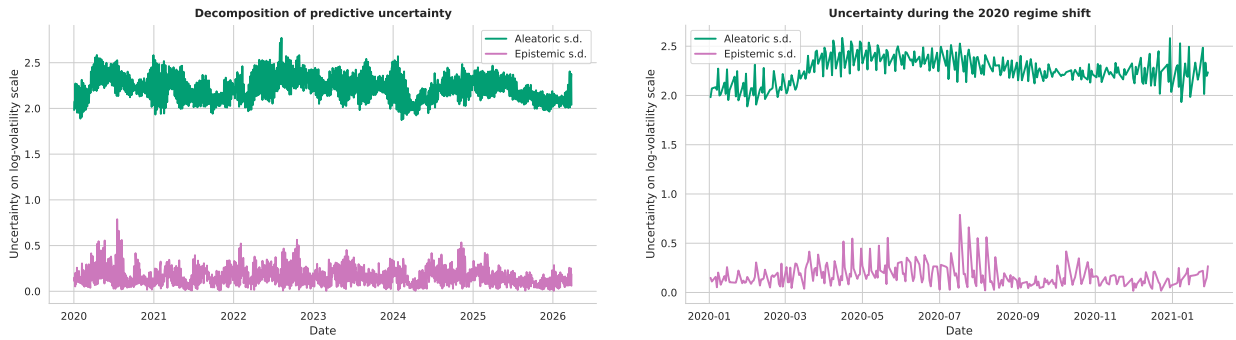
There are at least four reasons why the classical baselines remain superior in this experiment.

First, the target is noisy. Daily squared returns are highly variable and contain much more measurement noise than realized volatility computed from intraday data (Andersen et al., 2003; Barndorff-Nielsen and Shephard, 2002). When the signal-to-noise ratio is low, the main predictable component is often persistence at a few horizons, which HAR-style models capture directly.

Second, the information set is deliberately modest. The proposed architecture is designed so that it can eventually absorb richer covariates, OHLC range measures, realized volatility, macro-event embeddings, option-implied volatility, or text-derived sentiment, but those features are not present in the current data. The model is therefore asked to beat GARCH-like persistence with only daily cross-FX and calendar variables. That is a difficult benchmark.

Third, the sample is not especially large for deep learning. After lags and lookback windows, the Transformer trains on 3,958 sequences. That is substantial by econometric standards but small by the standards of modern deep sequence models. A compact architecture is therefore appropriate, yet compactness also limits expressiveness.

Fourth, the Gaussian head is convenient but likely too light-tailed. Even on the log-volatility scale, the empirical target is heavy-tailed and contains abrupt spikes. More flexible heads such as Student-$t$, skewed distributions, or mixture density networks (Bishop, 1994; Nix and Weigend, 1994) may capture these features better.




**(a)** Full test period.

**(b)** Zoom around the 2020 regime shift.

**Figure 7:** Aleatoric uncertainty remains the dominant component, while epistemic uncertainty spikes during periods of regime change and model disagreement. The 2020 sample shows a persistent elevation of aleatoric uncertainty and several sharp epistemic spikes.

## 7.2 Why the proposed framework is still useful

Losing the accuracy contest does not make the framework redundant. First, the model produces a coherent conditional law for next-day volatility rather than a single point forecast. In applications where downstream risk quantities are nonlinear in volatility, propagating a full density is often more natural than propagating only $\hat{\mu}_t$. Second, the decomposition in (29) adds information that classical conditional-variance models do not provide directly. A GARCH forecast already contains conditional heteroskedasticity, but it does not separate within-model market noise from cross-model disagreement in the way the ensemble does. Third, the architecture is extensible, if the information set is enlarged from $x_t$ to $(x_t, z_t)$, where $z_t$ may contain realized-volatility patches, macro-calendar tokens, or text embeddings, the same probabilistic head and the same uncertainty decomposition carry over immediately. The negative result reported here should therefore be interpreted as evidence about the specific low-information daily-data regime studied in this paper, not as a blanket rejection of probabilistic Transformers for FX volatility forecasting.

## 7.3 Practical implication for risk management

For the present daily USD/MXN setup, the best operational workflow is probably hybrid rather than doctrinaire. A classical model such as HAR-style Gaussian or EGARCH should remain the primary production forecast because it is sharper and better scoring in the experiments. The Transformer ensemble can then play a secondary role as a model-risk overlay, when epistemic uncertainty rises unusually fast, the risk manager receives a warning that the classical forecast may be operating outside familiar conditions. This is a concrete use-case where a model that does not win on NLL can still add value.

# 8 Conclusion

This paper examined next-day USD/MXN volatility forecasting as a distributional prediction problem, using a compact patch-based Transformer and a range of classical benchmark models.

The main empirical finding is that, for daily data and a noisy squared-return volatility proxy, simpler volatility models remain difficult to beat. In particular, the HAR-style Gaussian and GARCH-type benchmarks deliver stronger out-of-sample density forecasts on the main evaluation criteria.

At the same time, the Transformer results are still informative. The multivariate version improves on the univariate model, produces reasonably calibrated forecast intervals, and provides a useful uncertainty signal during periods of instability and regime change. The overall conclusion is therefore mixed but meaningful, the Transformer framework is viable for probabilistic volatility forecasting, but in this daily setting it does not yet outperform well-specified classical approaches.

# 9 Limitations and Future Work

A first limitation is the target variable itself. Daily squared returns are a very noisy proxy for latent volatility, so the forecasting problem is inherently difficult and may favour more parsimonious models. A second limitation is the information set, which in this study is restricted to daily market variables and simple calendar effects. This likely reduces the potential benefit of a more flexible architecture.

A more realistic setting for the Transformer would be one with richer inputs and stronger nonlinear structure. Future work could therefore use intraday realized volatility, OHLC-based range measures, macroeconomic event features, and text-based information from news or central-bank communication. It would also be useful to consider alternative distributional output layers and broader training across multiple currency pairs. In that kind of richer environment, the Transformer's flexibility and uncertainty estimates are more likely to provide a clear practical advantage.

# References


Andersen, T. G., Bollerslev, T., Diebold, F. X., and Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625.

Banco de Mexico (2022). Banco de mexico announces the results of the BIS triennial central bank survey of foreign exchange and over-the-counter derivatives markets in 2022. Press release. Accessed 2026-04-06. https://www.banxico.org.mx/publications-and-press/bis-triennial-central-bank-survey-of-foreign-exchange-and-over-the-counter-derivatives-markets-in-2022/results-of-the-bis-triennial-survey-2022.html.

Barndorff-Nielsen, O. E. and Shephard, N. (2002). Econometric analysis of realised volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B*, 64(2):253–280.

Bishop, C. M. (1994). Mixture density networks. Technical Report NCRG/94/004, Aston University.

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196.

datasets/exchange-rates contributors (2026). Exchange rates dataset. GitHub repository. Accessed 2026-04-06. https://github.com/datasets/exchange-rates.

Diebold, F. X., Gunther, T. A., and Tay, A. S. (1998). Evaluating density forecasts with applications to financial risk management. *International Economic Review*, 39(4):863–883.

Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of united kingdom inflation. *Econometrica*, 50(4):987–1007.

Federal Reserve Bank of St. Louis (2026). Mexican peso to one u.s. dollar spot exchange rate (DEXMXUS). FRED, Federal Reserve Bank of St. Louis. Accessed 2026-04-06. https://fred.stlouisfed.org/series/DEXMXUS.

Glosten, L. R., Jagannathan, R., and Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5):1779–1801.

Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378.

Hansen, P. R. and Lunde, A. (2005). A forecast comparison of volatility models: Does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7):873–889.

Jorion, P. (2007). *Value at Risk: The New Benchmark for Managing Financial Risk*. McGraw-Hill, New York, 3 edition.

Kendall, A. and Gal, Y. (2017). What uncertainties do we need in bayesian deep learning for computer vision? In *Advances in Neural Information Processing Systems*, pages 5574–5584.

Koenker, R. (2005). *Quantile Regression*. Cambridge University Press, Cambridge.

Koenker, R. and Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1):33–50.

Lakshminarayanan, B., Pritzel, A., and Blundell, C. (2017). Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, pages 6402–6413.

Lim, B., Arik, S. O., Loeff, N., and Pfister, T. (2021). Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4):1748–1764.

McNeil, A. J., Frey, R., and Embrechts, P. (2015). *Quantitative Risk Management: Concepts, Techniques and Tools*. Princeton University Press, Princeton, 2 edition.

Nelson, D. B. (1991). Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2):347–370.

Nguyen, Q. A., Guo, C., and Luk, W. (2025). Repurposing language models for fx volatility forecasting: A data-efficient and context-aware approach. In *Proceedings of the 6th ACM International Conference on AI in Finance*, pages 465–473.

Nie, Y., Nguyen, N. H., Sinthong, P., and Kalagnanam, J. (2023). A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations*.

Nix, D. A. and Weigend, A. S. (1994). Estimating the mean and variance of the target probability distribution. In *Proceedings of the 1994 IEEE International Conference on Neural Networks*, pages 55–60.

Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256.

Wen, Q., Zhou, T., Zhang, C., Chen, W., Ma, Z., Yan, J., and Sun, L. (2023). Transformers in time series: A survey. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*, pages 6778–6786.

Zeng, A., Chen, M., Zhang, L., and Xu, Q. (2023). Are transformers effective for time series forecasting? In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 11121–11128.

Zhao, L. and Yan, W. Q. (2024). Prediction of currency exchange rate based on transformers. *Journal of Risk and Financial Management*, 17(8):332.

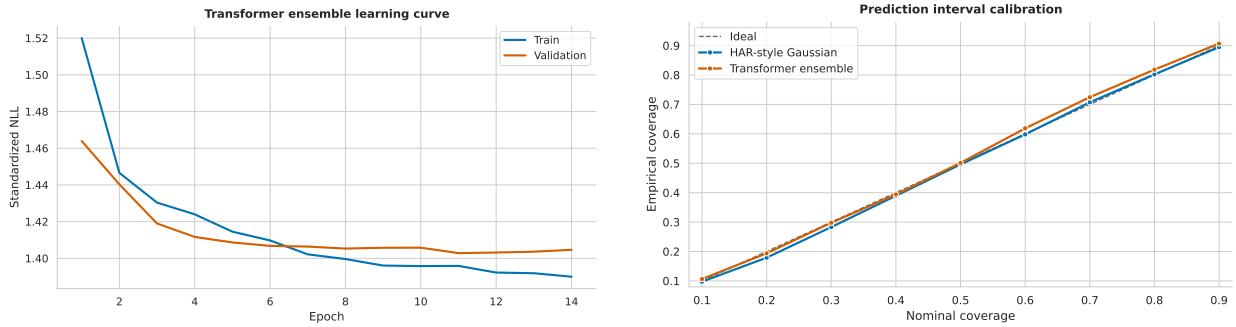
# A Appendix

## A.1 Data Summary

**Table 4:** Dataset summary and Transformer configuration.

| Panel A: Data summary | | Panel B: Transformer configuration |
|---|---|---|
| Source | FRED DEXMXUS via public GitHub mirror | Lookback window: 40 trading days |
| Span | 2000-01-03 to 2026-03-27 | Patch size: 5 days |
| Aligned daily observations | 6,577 | Number of patches: 8 |
| Target | $y_t = \log(r_t^2 + \varepsilon)$ | Input dimension: 23 features/day |
| Additional currencies | BRL, CAD, EUR, ZAR, JPY, KRW, CNY | Encoder layers: 1 |
| Train / validation / test sequences | 3,958 / 998 / 1,558 | Attention heads: 4 |
| Calendar variables | Day-of-week dummies | Embedding dimension: 16 |
| HAR variables | Lagged daily, weekly, monthly averages | Feed-forward width: 32 |
| | | Optimizer: AdamW, learning rate $6 \times 10^{-4}$ |
| | | Early stopping patience: 4 epochs |
| | | Ensemble size: 3 members |

## A.2 Training Evaluation




**(a)** Validation-based early stopping is triggered quickly.

**(b)** Empirical interval coverage remains close to nominal.

**Figure 8:** Training dynamics and interval calibration for the Transformer ensemble. The compact model does not require long training and produces intervals that are slightly conservative but close to the ideal coverage line.

## A.3 Additional Results by Volatility Regime

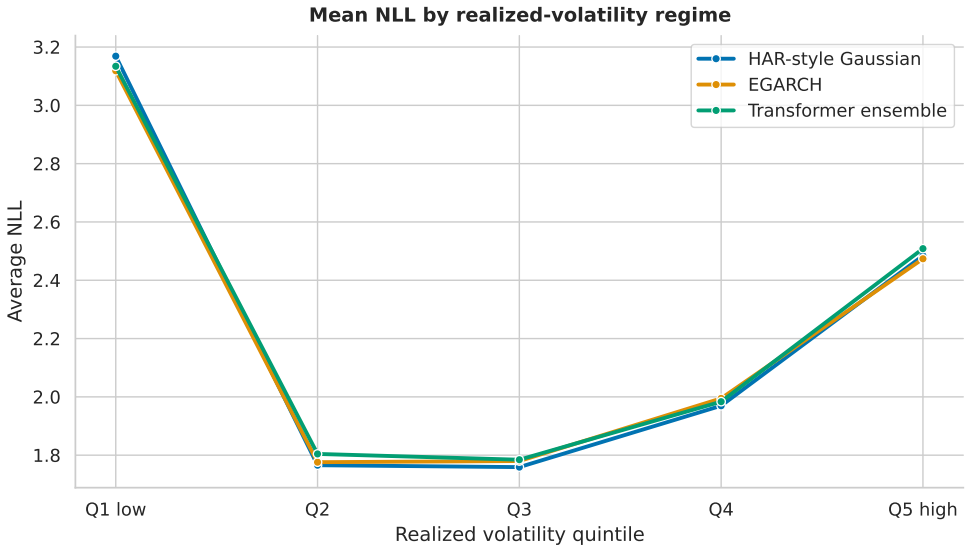



**Figure 9:** Mean NLL stratified by realized-volatility quintile. HAR-style Gaussian is strongest across the middle regimes, EGARCH is marginally best in the calmest and most turbulent quintiles, and the Transformer ensemble remains competitive but does not take the lead in any regime.